THE SPECTRAL STRUCTURE OF FRACTIONAL FIELDS

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Abstract. For primitive-root primes ($\mathrm{ord}_p(b) = p-1$), we prove that the cross-alignment matrix, reordered by discrete-logarithm offset, is a circulant whose eigenvalues are the Fourier transform of the digit-equality autocorrelation. For primes with exactly two cosets and zero within-coset autocorrelation at nonzero lags, the spectrum has two eigenvalue levels determined by the cross-coset match count. The multi-coset case is described observationally.

1. Introduction

The cross-alignment matrix $\mathbf{A}(p)$ introduced in [1] is a symmetric, positive semidefinite matrix on the fractions of a prime p in base b, whose (i, j) entry is the digit-equality proportion between i/p and j/p. For digit-partitioning primes ($p \leq b+1$), the matrix is the identity, every fraction is a stranger to every other, and the eigenvalue spectrum is trivially flat. For non-digit-partitioning primes, the matrix is no longer the identity. Its eigenvalue spectrum encodes the interference geometry of the field, but the spectrum was not given a closed form in [1].

This paper identifies the eigenvalues. The main result (Theorem 2) is that for primes p where b is a primitive root modulo p, the matrix $\mathbf{A}(p)$, after a discrete-logarithm reordering of its rows and columns, becomes a circulant matrix whose first row is the cyclic autocorrelation of the base repetend $1/p$, divided by the cycle length $L = p - 1$. Standard properties of circulants [2] then identify the eigenvalues as the discrete Fourier transform of the first row. The eigenvalue spectrum of $\mathbf{A}(p)$ is a character expansion of the repetend's autocorrelation.

For primes p where b is not a primitive root, the fractions partition into $C = (p-1)/L$ orbits of $\langle b \rangle$, and the matrix decomposes into within-orbit and between-orbit blocks. We treat the two-orbit case in detail (Theorem 4) and show that under the natural sufficient conditions, which are satisfied at $p = 13$ in base 10, the spectrum collapses to two

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eigenvalue levels determined by an involution. The general multi-orbit case is described observationally in §5.

The connection to harmonic analysis on finite groups is explicit: the eigenvalues are inner products of the autocorrelation function with the additive characters of $\mathbb{Z}/L\mathbb{Z}$. This identifies the eigenvalue spectrum as a Fourier transform on a finite cyclic group, which is the natural starting point for the harmonic analysis pursued in the next several papers.

2. The Circulant Theorem

Definition 1. The *cyclic autocorrelation* of the base repetend $(d_0, d_1, \ldots, d_{L-1})$ of $1/p$ is the sequence

$$R(\ell) = |\{j \in \{0, \ldots, L-1\} : d_j = d_{(j+\ell) \bmod L}\}|, \qquad \ell = 0, 1, \ldots, L-1.$$

Note that $R(0) = L$ (every position matches itself) and $R(\ell) \geq 0$ for all ℓ.

Theorem 2. *Let p be a prime for which b is a primitive root ($\mathrm{ord}_p(b) = p-1$). Then the cross-alignment matrix $\mathbf{A}(p)$, reordered by discrete-logarithm offset, is a circulant matrix with first row*

$$\left(\frac{R(0)}{L}, \frac{R(1)}{L}, \frac{R(2)}{L}, \ldots, \frac{R(L-1)}{L}\right).$$

Its eigenvalues are given by the discrete Fourier transform [2]*:*

$$\lambda_j = \frac{1}{L}\sum_{\ell=0}^{L-1} R(\ell)\,\omega^{j\ell}, \qquad j = 0, 1, \ldots, L-1,$$

where $\omega = e^{2\pi i/L}$.

Proof. When b is a primitive root modulo p, every fraction k/p is a cyclic rotation of $1/p$ in the sense that the repetend of k/p in base b is the repetend of $1/p$ shifted by $\log_b k$ positions modulo L (where $\log_b$ denotes the discrete logarithm with respect to b modulo p). Ordered by rotation offset, the fractions are $1/p,\ b/p,\ b^2/p,\ \ldots,\ b^{L-1}/p$ (each reduced modulo p).

In this ordering, the (i, j) entry of $\mathbf{A}(p)$ is the digit-equality proportion between b^i/p and b^j/p, which is the proportion of positions in the repetend of $1/p$ where the digit at position ℓ agrees with the digit at position $\ell+(j-i) \bmod L$. By the definition of R, this proportion is $R(j-i \bmod L)/L$. The (i, j) entry depends only on the difference $j-i$ modulo L, so the matrix is circulant with first row $(R(0)/L,\ R(1)/L,\ \ldots,\ R(L-1)/L)$.

The eigenvalues of a circulant are the discrete Fourier transform of its first row [2], which yields the stated formula. □

Corollary 3. *If p is digit-partitioning ($p \leq b+1$), every eigenvalue of $\mathbf{A}(p)$ equals 1.*

Proof. By [1], $\mathbf{A}(p) = I_{p-1}$ for digit-partitioning p, so the spectrum is the constant sequence 1. For primitive-root digit-partitioning primes, the same conclusion follows from Theorem 2: the digit-partitioning condition forces $R(\ell) = 0$ for $\ell \neq 0$ (no two distinct positions of the repetend share a digit), and the DFT of $(L, 0, 0, \ldots, 0)/L$ is the constant sequence 1. □

3. Worked Example: $p = 7$ in Base 10

In base 10, the prime $p = 7$ is digit-partitioning ($p \leq b+1 = 11$) and primitive-root ($\mathrm{ord}_7(10) = 6$). The repetend of $1/7$ is 142857, of length $L = 6$. By Theorem 2, $\mathbf{A}(7)$ in the discrete-logarithm ordering is a 6×6 circulant whose first row is $(R(0)/6, R(1)/6, \ldots, R(5)/6)$.

The autocorrelation values are computed by direct comparison of 142857 with its rotations. At lag $\ell = 1$ the rotated sequence is 428571, and the position-wise comparison (1:4, 4:2, 2:8, 8:5, 5:7, 7:1) yields zero matches. The same calculation at every lag $\ell \in \{1, 2, 3, 4, 5\}$ gives zero matches. Hence $R(0) = 6$ and $R(\ell) = 0$ for $\ell \neq 0$, and the first row of the circulant is $(1, 0, 0, 0, 0, 0)$.

The DFT of this row is the constant sequence $(1, 1, 1, 1, 1, 1)$. All six eigenvalues of $\mathbf{A}(7)$ equal 1, in agreement with the digit-partitioning case of Corollary 3.

4. The Two-Orbit Case

When $\mathrm{ord}_p(b) = L < p-1$, the fractions partition into $C = (p-1)/L$ orbits of $\langle b \rangle$ in $(\mathbb{Z}/p\mathbb{Z})^*$. For $C = 2$, the cross-alignment matrix has additional structure that we make precise below.

Theorem 4. *Let p be a prime with exactly two orbits of $\langle b \rangle$ in $(\mathbb{Z}/p\mathbb{Z})^*$, each of size $L = (p-1)/2$. Suppose*

1. *the within-orbit autocorrelation $R_C(\ell)$ (the digit-equality count between rotations within the same orbit) satisfies $R_C(\ell) = 0$ for $\ell \neq 0$, and*
2. *each fraction in one orbit matches exactly one fraction in the other orbit at exactly m positions, with all other cross-orbit pairs matching at zero positions.*

Then the eigenvalues of $\mathbf{A}(p)$ *are*

$$1 + \frac{m}{L} \quad \textit{and} \quad 1 - \frac{m}{L},$$

each with multiplicity $L = (p-1)/2$.

Proof. Under these conditions, every diagonal entry of $\mathbf{A}$ is 1, every within-orbit off-diagonal entry is 0 (from condition (1)), and every cross-orbit off-diagonal entry is either m/L (for the unique partner) or 0 (otherwise). We can therefore write

$$\mathbf{A}(p) = \mathbf{I} + \frac{m}{L}\,\mathbf{P},$$

where $\mathbf{P}$ is the permutation matrix that pairs each fraction with its unique cross-orbit partner. Since the pairing is symmetric, $\mathbf{P}$ is an involution ($\mathbf{P}^2 = \mathbf{I}$), so its eigenvalues are ± 1, each with multiplicity equal to half its dimension, i.e., multiplicity L each. The eigenvalues of $\mathbf{A}$ follow as $1 \pm m/L$. □

Proposition 5. *The hypotheses of Theorem 4 are satisfied at* $p = 13$ *in base* 10*, with* $m = 2$ *and* $L = 6$*. The eigenvalues of* $\mathbf{A}(13)$ *are* $4/3$ *and* $2/3$*, each with multiplicity* 6*.*

Proof. The order of 10 modulo 13 is 6, so $L = 6$ and $C = 2$. The two orbits of $\langle 10 \rangle$ in $(\mathbb{Z}/13\mathbb{Z})^*$ are $\{1, 10, 9, 12, 3, 4\}$ and $\{2, 7, 5, 11, 6, 8\}$. The repetends within the first orbit are the rotations of 076923, and within the second orbit they are the rotations of 153846.

Within-orbit comparison: comparing 076923 with each of its rotations 769230, 692307, 923076, 230769, 307692 position by position gives zero matches in every case. Hence $R_C(\ell) = 0$ for $\ell \neq 0$, verifying condition (1).

Cross-orbit comparison: comparing $1/13 = 076923$ with each member of the second orbit gives the following position-wise match counts:

partner	matches
$2/13 = 153846$	0
$5/13 = 384615$	0
$6/13 = 461538$	0
$7/13 = 538461$	0
$8/13 = 615384$	0
$11/13 = 846153$	2

The unique partner of $1/13$ in the other orbit is $11/13$, matching at exactly 2 positions (positions 3 and 6). The same pattern holds for every other fraction in the first orbit by cyclic symmetry. Every member

of the first orbit is a rotation of 076923, and every member of the second orbit is a rotation of 153846. Rotating both strings by the same amount preserves the number of aligned matches at corresponding positions, so the comparison pattern for any fraction in the first orbit against the second orbit is obtained by rotating the table above. Hence each fraction has a unique cross-orbit partner matching at exactly 2 positions and matches every other cross-orbit fraction at 0 positions. This verifies condition (2) with $m = 2$.

By Theorem 4, the eigenvalues of $\mathbf{A}(13)$ are $1 + 2/6 = 4/3$ and $1 - 2/6 = 2/3$, each with multiplicity 6. □

Whether the hypotheses of Theorem 4 hold for every two-orbit prime, or only for a subfamily that includes $p = 13$, is not determined here.

5. The General Multi-Orbit Case

For primes with $C = (p-1)/L > 2$ orbits, the cross-alignment matrix has a block decomposition that we describe but do not prove in full generality.

Proposition 6. *For any prime p with C orbits of $\langle b \rangle$ in $(\mathbb{Z}/p\mathbb{Z})^*$, each of size L, the cross-alignment matrix admits the decomposition*

$$\mathbf{A}(p) = \mathbf{W}(p) + \mathbf{X}(p),$$

where $\mathbf{W}(p)$ is the block-diagonal within-orbit contribution (a direct sum of C circulant $L \times L$ matrices, one per orbit, each governed by the within-orbit autocorrelation) and $\mathbf{X}(p)$ is the block-off-diagonal between-orbit contribution (whose (i, j) block records the cross-orbit alignments between orbit i and orbit j).

Proof. The block partition follows from grouping the rows and columns of $\mathbf{A}(p)$ by orbit. Within each orbit of size L, the rotation argument of Theorem 2 applies verbatim, since fractions within the same orbit are rotations of one another, so each diagonal block is a circulant whose first row is $R_C(\ell)/L$ for that orbit. The cross-orbit blocks contain the digit-equality proportions between fractions of different orbits and have no cyclic structure in general. □

The within-orbit eigenvalues are the DFT of $R_C(\ell)/L$ for each orbit, by Theorem 2 applied within the orbit. The cross-orbit blocks add a perturbation whose eigenvalues depend on the joint structure of all C orbits.

Computation for all primes $p \leq 37$ in base 10 shows that the spectrum of $\mathbf{A}(p)$ is determined by the within-orbit autocorrelations $R_C(\ell)$ together with the cross-orbit alignment data, in the sense that two

primes with the same orbit structure and the same alignment data have the same spectrum. A closed-form expression for the multi-orbit eigenvalues in terms of these data is not given here.

6. Connection to Harmonic Analysis

The DFT that produces the eigenvalues is a sum over characters of $\mathbb{Z}/L\mathbb{Z}$:

$$\lambda_j = \frac{1}{L}\sum_{\ell=0}^{L-1} R(\ell)\,\chi_j(\ell),$$

where $\chi_j(\ell) = e^{2\pi i j\ell/L}$ is the j-th character. The eigenvalue λ_j is therefore the inner product of the autocorrelation function with the character χ_j.

This identifies the eigenvalue spectrum of the cross-alignment matrix as a *character expansion* of the repetend's autocorrelation, in the sense of harmonic analysis on finite abelian groups [3]. The digit-partitioning property corresponds to a flat character expansion (all coefficients equal). Non-DP primes exhibit character-dependent structure.

For primitive-root primes, where $L = p - 1$, the additive characters of $\mathbb{Z}/L\mathbb{Z}$ used here are not the Dirichlet characters of $(\mathbb{Z}/p\mathbb{Z})^*$, but the two character groups are isomorphic via the discrete logarithm: the map $\log_b : (\mathbb{Z}/p\mathbb{Z})^* \to \mathbb{Z}/L\mathbb{Z}$ intertwines a multiplicative character of $(\mathbb{Z}/p\mathbb{Z})^*$ with an additive character of $\mathbb{Z}/L\mathbb{Z}$. The eigenvalues of $\mathbf{A}(p)$ can therefore be re-expressed as evaluations of Dirichlet characters modulo p on the autocorrelation pulled back through the discrete logarithm. The Dirichlet character framework [4] provides the analytic machinery for relating these character sums to L-functions, and is the natural setting in which the next several papers in the chain are developed.

7. Remarks

The autocorrelation in the primitive-root case. For primitive-root primes, the cyclic autocorrelation $R(\ell)$ determines the full eigenvalue spectrum (Theorem 2). For multi-coset primes, the within-coset autocorrelation alone does not suffice: the between-coset cross-correlation data is also needed (as seen at $p = 13$ in base 10, where $R(\ell) = 0$ for $\ell \neq 0$ yet the spectrum is not flat).

From spectra to L-functions. The expression $\lambda_j = L^{-1}\sum R(\ell)\chi_j(\ell)$ resembles a Gauss-type sum. If the autocorrelation $R(\ell)$ can be expressed in terms of the Legendre symbol or other multiplicative characters, the eigenvalues become evaluations of Dirichlet L-functions at

specific points. Whether this connection yields non-trivial analytic information about the fractional field is open.

References

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